

I/L ①② 0/L

shallow network 1-2 hidden

linearly, mid
work

moderate

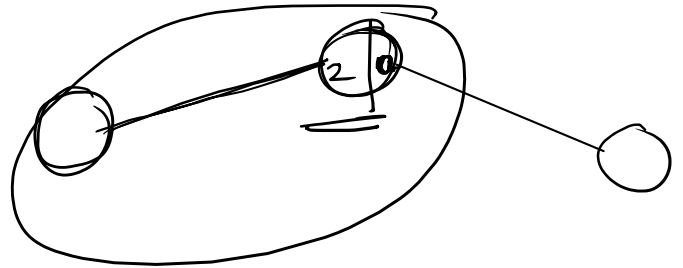
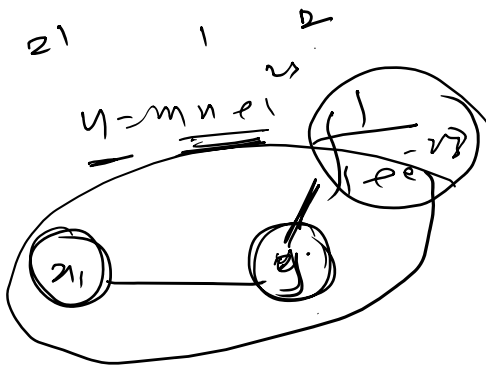
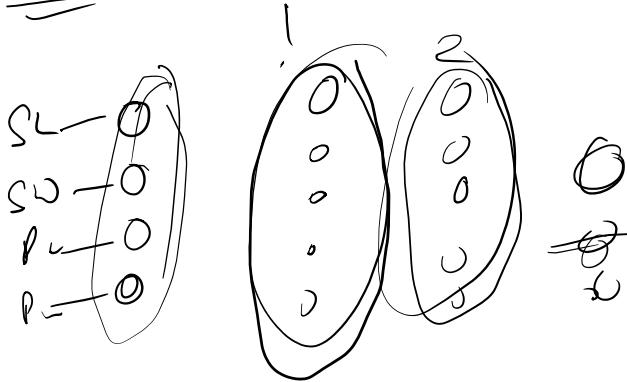
2-5

text, image

deep

10+ layers

1c+15, audio
work



build the model $f_t = ()$

compile x

fit $\longrightarrow ()$

iris data - 150 samples
80% train 20% test

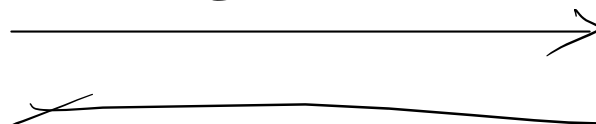
$$\frac{150 \times 80}{100} = 120$$

$(x_{train}, y_{train}) \rightarrow \underline{120}$

batch = 5

24 24 24 24 24

epoch w.



0 [0 0 0]

$\xrightarrow{\quad}$ 0 $[000]$
 $\xleftarrow{\quad}$ coal-tune, optuna, scikit-learn $[010]$
 2 $[001]$

$x = d([\quad])$
 $y = d([\quad]) (-1, 1)$

$[\quad]$

$[000000]$
 $[100000]$

$0 \rightarrow [\quad]$

$0 \rightarrow [2^2 \ 2^1 \ 2^0]$

$1 \rightarrow$
 $2 \rightarrow$

$000 \rightarrow 0 \rightarrow [000]$
 $001 \rightarrow 1 \rightarrow [001]$
 $010 \rightarrow 2 \rightarrow [010]$

000
 1